Topology-Aware Hierarchical Mamba for Salient Object Detection in Remote Sensing Imagery

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***Abstract*—Salient object detection (SOD) plays a crucial role in the intelligent interpretation of remote sensing tasks. Significant advancements have been made with SOD methods based on convolutional neural networks (CNNs) and Transformers. However, the structural differences and complex topological relationships presented by diverse remote sensing scenes often limit SOD performance. To address these challenges, this study proposes a novel method for SOD in remote sensing images (RSI) called topology-aware hierarchical Mamba network (THMNet), which fully utilizes the Mamba's advantages in global modeling and linear computation. Specifically, a grouped Mamba topology enhancement module is designed to extract high-level semantic features and improve the global representation of structural information for salient objects. Furthermore, a hierarchy Mamba segmentation optimization (HMSO) module is introduced, which employs the prior guidance to adaptively focus on the foreground and background at different levels. This module reduces the processing of noncritical information within complex topological relationships and enhances the capture and depiction of intricate objects. Finally, experimental results demonstrate that our method outperforms mainstream CNNs- and Transformers-based SOD techniques, effectively detecting salient objects in complex backgrounds. The code is available at: https://github.com/LazyShark2001/THMNet**

***Index Terms*—Mamba, remote sensing images (RSI), salient object detection (SOD), topology enhancement.**

I. Introduction

In RECENT years, significant progress has been made in salient object detection (SOD) in the field of natural image analysis [1], [2]. Based on human visual attention mechanisms, SOD techniques in remote sensing imagery aim to accurately locate and segment the most visually prominent objects. This supports traditional tasks, such as object segmentation [3], image quality assessment [4], and visual tracking [5], while also advancing the application of remote sensing data in intelligent analysis and change detection, thereby demonstrating considerable practical value and research significance. However, the unique characteristics of remote sensing images (RSIs)—such as complex geospatial information, diverse object scales, and distinct viewing angles—limit the application of existing SOD methods designed for natural images. Consequently, researchers are focused on developing detection techniques specifically tailored for remote sensing scenarios to more effectively identify salient regions within these images [6], [7], [8].

Traditional SOD methods primarily rely on manually designed features, including feature analysis and fusion [9], [10], sparse representation [11], and color information [12]. Although these approaches have shown certain effectiveness in specific tasks, they often suffer from low computational efficiency and struggle to achieve the desired accuracy in practical applications, particularly when adapting to various scenes, resulting in poor robustness.

With the rapid advancement of deep learning, convolutional neural networks (CNNs) have established a dominant position in the field of computer vision, significantly advancing SOD in remote sensing, particularly in areas such as hyperspectral image classification and semantic segmentation of optical RSIs [13]. Huo et al. [14] proposed a new framework designed to extract global information and long-range dependencies. This framework employs a Transformer-based backbone network to capture these dependencies and utilizes a global guidance branch to learn contextual relationships, yielding promising results. Zheng et al. [15] developed a cross-scale interaction network that employs dilated convolutions to expand the receptive field while utilizing skip connections to capture a richer array of multiscale contextual information, resulting in a marked improvement in visual performance. Li et al. [16] designed a lightweight network based on semantic matching and edge alignment. This network uses a simplified CNN backbone to extract features and dynamically activate salient object locations, achieving efficient SOD. In addressing the RSI-SOD task, Li et al. [17] introduced the LVNet model, which employs an L-shaped dual-stream pyramid module to extract multiscale features from images at various resolutions, in

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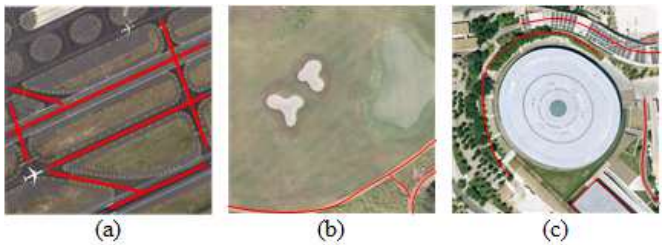


Fig. 1. (a) Complex spatial topological relationship formed by multiple aircraft taxiways, parking spaces, and markings. (b) Image encompasses diverse land features, with roads displaying unique geometric properties regarding orientation, curvature, and boundary connectivity, collectively giving rise to a complex morphological topology. (c) Building in the center of the image displays a concentric ring structure, showcasing multiple hierarchical layers, which constitute a significant topological feature.

conjunction with a V-shaped nested encoder–decoder module to highlight salient targets while suppressing background noise. Zhang et al. [18] proposed the multitask framework DAFNet, which synchronously infers salient maps and edge maps through edge supervision and attention information transfer from shallow to deep layers, thereby further enhancing edge information. Moreover, some studies have enhanced the networks' ability to distinguish between foreground and background in RSIs by integrating edge information. This approach demonstrates strong robustness when dealing with complex backgrounds, effectively mitigating background interference and improving the accuracy of object detection [19], [20], [21], [22].

Despite the effectiveness of the aforementioned methods in addressing specific problems, they still exhibit significant limitations when dealing with the complex and variable backgrounds present in RSIs. Such background factors often interfere with the segmentation and identification of salient objects, leading to decreased detection accuracy. As shown in Fig. 1, the background of RSIs contains complex topological structures, which are mainly reflected in two aspects. The first is the shape topology within individual objects, such as whether roads are continuous, whether building boundaries are closed, or whether rivers remain connected. This aspect emphasizes the structural integrity of a single object. If shape topology is not sufficiently modeled, the segmentation results may exhibit fragmented, disconnected, or broken object boundaries. The second is the semantic topology between different objects, such as the adjacency between buildings and roads or vegetation, the intersections of roads and rivers, and the separation of foreground objects from background regions. This aspect highlights the global organizational structure of foreground and background. Inadequate modeling of semantic topology may lead to adhesion between foreground and background or the erroneous merging of adjacent objects. These multilevel topological relationships arise from diverse land-cover types, irregular object shapes, and intricate spatial interactions among different elements, which not only increase the difficulty of distinguishing foreground from background but also exacerbate the visual confusion between salient objects and their surroundings. In optical RSIs, background features are typically dispersed, irregular, and challenging to recognize through simple regularization patterns. If the network's receptive field is too small, it struggles to capture the global features of these varied topological structures, thus adversely affecting the accurate classification of foreground and background. This imposes higher demands on remote sensing SOD, necessitating that networks possess enhanced spatial information comprehension abilities and can capture subtle details and complex relationships within the background across different scales. Achieving this often relies on deep and complex network architectures capable of enlarging the receptive field and strengthening feature extraction capabilities. Such an approach, however, incurs substantial computational costs, as complex networks typically require significant computational resources and memory, particularly in the context of processing large-scale remote sensing datasets. This presents a formidable challenge for hardware resources and computational efficiency. Therefore, how to maintain high accuracy in SOD in complex backgrounds while controlling the usage of computational resources has become a pressing issue in the field of remote sensing SOD.

To address the challenges of SOD in remote sensing imagery, this study proposes a remote sensing SOD network based on topology-aware hierarchical Mamba. This network employs a hierarchical strategy to capture both the shape topology and spatial semantic topology of land features in complex backgrounds. First, the group Mamba topology enhancement (GMTE) module performs global directional modeling to enhance the shape-topological features of targets. This step is crucial, as accurate modeling of object shape topology provides a reliable foundation for parsing the spatial topological relationships among different targets. Subsequently, guided by the shape priors from GMTE, the hierarchy Mamba segmentation optimization (HMSO) module performs hierarchical optimization of the foreground, background, and overall scene, further refining the spatial semantic topological representation between targets. This cross-level feature coordination significantly enhances target discriminability. Specifically, the network employs an encoder–decoder architecture, in which the encoder uses a mainstream feature extraction backbone for the encoder. Building upon this, two key modules are designed to further enhance object recognition accuracy. First, we designed a GMTE module, which enhances the global shape topology by integrating multiscale feature maps and performing directional modeling along four orientations. This module captures the boundary connectivity, internal structure, and overall morphology of targets, providing precise shape priors for subsequent spatial semantic topology modeling. Subsequently, we developed a HMSO module. Guided by GMTE, HMSO decouples foreground, background, and global features, and computes pixelwise correlations across different regions through multiple Mamba units to differentiate foreground from background and refine intertarget spatial semantic topology. This hierarchical design not only improves the clarity of target boundaries but also effectively suppresses the interference of complex backgrounds in SOD. Through the hierarchical Mamba structures, the HMSO module achieves information complementarity and functional coordination between foreground and background

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features, further optimizing the discriminative capability for the target and background in the image. This structure not only enhances the clarity of object boundaries but also effectively suppresses the interference from complex backgrounds in SOD.

The contributions of this research are as follows.

1) This study introduces a topology-aware hierarchical Mamba network (THMNet) for remote sensing SOD to tackle the challenges posed by optical remote sensing image SOD (ORSI-SOD). THMNet's decoder utilizes distinct Mamba structures for foreground and background regions and adopts a progressive top–down refinement strategy to incrementally improve the segmentation quality across feature levels.
2) To address the issues of incomplete target segmentation and insufficient capability in capturing topological relationships, we introduce the GMTE module. GMTE performs global directional modeling on the highest level semantic feature maps, capturing both global semantics and multiscale contextual relationships. From a holistic perspective, it finely characterizes the shape topology of targets, thereby enhancing the network's deep understanding of overall topological relationships.
3) We introduce the HMSO module to perform refined segmentation of multiscale features. Guided by the shape-topology information provided by GMTE, HMSO conducts focused spatial-topology modeling at the foreground, background, and global levels. This ensures that high-level features effectively guide the refinement of lower level, multiscale segmentation, enabling the network to more accurately delineate target boundaries and improve overall segmentation performance.

The remainder of this article is organized as follows. Section II reviews related work. Section III presents our proposed framework and its components, including the designed loss function. Section IV provides comprehensive experimental results and evaluates the contribution of each component. Section V offers further discussion of the method. Finally, Section VI concludes this article.

II. Related Works

A. SOD in Optical Remote Sensing Imagery

In recent years, with the rapid advancement of the optical RSI SOD (RSI-SOD), researchers have proposed various innovative models aimed at enhancing the accuracy and robustness of SOD in RSIs. For example, Li et al. [6] proposed a multicontent complementary network that employs a multicontent complementary module between the encoder and decoder. The MCCM module operates on multiscale features, leveraging the complementary information among foreground features, edge features, background features, and global image-level features. An attention mechanism is used to highlight the salient regions of RSI features across different scales. Sun et al. [23] developed a channel-enhanced reshaping network that improves the feature representation and localization capabilities of salient objects through channel enhancement modules and redefined feature modules. Zeng et al. [22] proposed an adaptive edge-aware semantic interaction network that effectively enhances the edge details of salient objects using a difference-aware mechanism and multiscale feature extraction modules. Di et al. [24] introduced WeightNet, which employs a two-level design that combines multiscale weighted feature aggregation modules with multidimensional feature guidance modules to achieve efficient fusion of multiscale features. Quan et al. [25] designed a global semantic enhancement block that suppresses background interference and accurately locates salient objects through interscale fusion and semantic reasoning modules, yielding promising results. Shi et al. [26] proposed a robust adversarial defense algorithm that significantly improves model performance under adversarial noise using multimodal data fusion and global context denoising modules. Yan et al. [27] introduced a global–local semantic interaction network that integrates Transformer and CNN. This network enhances the feature interaction through global–local fusion modules to better capture contextual information. Zhao et al.'s [28] RAGRNet effectively captures salient objects in complex RSIs using multigranularity boundary focusing modules and recursive adaptive graph reasoning modules. Huo et al. [14] developed a Transformer-based GMANet that strengthens edge details and global information through multiscale attention modules and global guidance branches. Zheng et al. [29] proposed a boundary-aware network that enhances the clarity and structural integrity of salient object edges through boundary- and structure-aware decoders. In addition, Zheng et al. [15] introduced a cross-scale interaction network that improves adaptability to objects of varying scales and shapes through a Forked skip connection module (FSCM) and an expanded-integrated module (EIM). These research methods have significantly advanced the performance of SOD in remote sensing imagery through various innovative techniques, including feature enhancement, attention mechanisms, and multiscale fusion, providing reliable support for detecting objects in complex-background conditions. However, many of the aforementioned approaches rely on highly coupled modules when addressing the foreground–background discrimination problem in complex backgrounds. They often fail to effectively distinguish between foreground and background features; instead, they tend to treat these features indiscriminately without utilizing multiple decoupled branches to separately enhance the features of each, thereby neglecting to emphasize the contrasting information between the two.

B. State-Space Models (SSMs) and Mamba Architecture

Recently, Gu and Dao [30] proposed a data-dependent SSM layer and developed a general language model backbone called Mamba. This model outperformed various Transformer architectures across diverse real-world datasets and demonstrated linear scalability for longer sequences. Following this, Visual Mamba [31] and Vision Mamba [32] extended the Mamba architecture to 2-D image data, achieving remarkable results in multiple computer vision tasks. This advancement has inspired a series of pioneering works, driving the application of the Mamba architecture in various image processing domains. Among these, Chen et al.

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[33] introduced three models—MambaBCD, MambaSCD, and MambaBDA—tailored for binary change detection, semantic change detection, and assessment of building damage, respectively. All these models utilize Visual Mamba as the encoder, effectively learning global spatial contextual information and yielding impressive results. Chen et al. [34] developed the SSM-based RSMamba, which integrates a dynamic multipath activation mechanism to enhance modeling capabilities for noncausal sequences, demonstrating the superior performance in multiple remote sensing image classification tasks. Zhao et al. [35] proposed an RSM architecture specifically designed for dense prediction in high-resolution remote sensing images. It employs an omnidirectional selective scanning module to achieve global modeling from various directions, resulting in significant improvements in image segmentation and change detection tasks. In addition, Fu et al. [36] introduced the memory-efficient SSUMamba architecture, with the SSCS Mamba module focusing on hyperspectral image (HSI) denoising. This module captures long-range spatial–spectral dependencies through bidirectional SSM, and incorporates 3-D convolution to enhance spatial–spectral modeling. Liao et al. [37] constructed a lightweight model called LightM-UNet, which uses residual Mamba layers to strengthen long-range modeling capabilities, achieving efficient segmentation and performing exceptionally well across various 2-D and 3-D datasets. Yue and Li [38] presented MedMamba, a specialized architecture for medical image classification. This architecture integrates SS-Conv-SSM mixed modules to effectively capture long-range dependencies while reducing computation through grouped convolution and channel shuffle techniques. Furthermore, Li et al. [39] developed the MambaDFuse model which combines CNNs and Mamba for feature extraction, and designed a dual-phase feature fusion module to integrate complementary multimodal information, significantly enhancing the quality of fused images. These studies collectively indicate that the Mamba architecture exhibits substantial potential and outstanding performance in visual tasks. Models based on the Mamba architecture not only achieve robust generalization across a variety of visual scenarios but also adapt effectively to varying data scales and task requirements.

III. Proposed Method

A. Network Overview

SOD in remote sensing imagery faces significant challenges, primarily due to the rich details and diverse scenes present in high-resolution images. The complex details and textures in high-resolution images increase the presence of distracting elements in the background, making it more difficult to distinguish between foreground and background features. When a network has a limited receptive field, it often struggles to adequately extract long-range spatial dependencies under the influence of complex backgrounds. This ultimately leads to misclassification of foreground and background, especially in remote sensing imagery with intricate topological structures. Moreover, remote sensing images typically contain large-scale datasets, so it often takes lots of computing resources to construct network architectures that possess both a large receptive field and robust spatial understanding capabilities.

To address these challenges, this study proposes a hierarchical Mamba topology enhanced network for remote sensing SOD, referred to as THMNet. This network adopts a classic encoder–decoder architecture, as illustrated in Fig. 2. In the first stage of the network, the encoder generates multiscale image features, laying the groundwork for subsequent detection tasks. Specifically, the PVTv2 architecture is used as the encoder. PVTv2 is composed of four stages, each including a patch embedding layer and several Transformer blocks. In PVTv2, the image is divided into 4×4 spatial patches, which are flattened and passed through Transformer blocks after positional encoding. Following a series of feature extraction and spatial transformation processes, the encoder outputs four multiscale feature maps with dimensions of 64, 128, 320, and 512, corresponding to downsampling factors of 4, 8, 16, and 32, respectively. After the encoder generates image features, the second stage of the network employs the GMTE module, performing multiscale feature fusion using the encoder outputs. This process optimizes object shape topology while capturing global information, thereby improving the quality of the saliency prediction map. Specifically, GMTE performs cumulative contextual modeling on multiscale semantic feature maps through directional scanning of the Group Mamba units, enhancing target boundary continuity, internal connectivity, and overall shape topology. This approach enables the network to develop a refined understanding of object shapes at a global scale, rather than merely capturing local pixel relationships. In the third stage, we adopt a progressive approach to refine features from the top of the encoder. Specifically, at each feature level, HMSO employs three separate Mamba heads to perform focused spatial-topology modeling on the foreground, background, and overall features independently. This decoupled design allows each branch to capture its unique spatial dependencies and topological patterns, avoiding mutual interference between foreground and background information, and thereby enabling the network to better distinguish salient targets from the background. Following this, predictions are made at each feature level with the inclusion of side supervision. Through the iterative optimization of HMSO, shallow feature maps are enriched with semantic information, enabling the network to capture object boundaries more accurately and ensuring high-quality segmentation outcomes at shallow layers.

B. Group Mamba Topology Enhancement

In high-resolution remote sensing images, abundant fine-grained details and complex shape topologies often lead to incomplete target segmentation, while background interference complicates the distinction between foreground and background, thereby affecting the accuracy of overall topological relationships. This issue is particularly pronounced in complex scenes. Although self-attention mechanisms can construct fully connected pixelwise semantic graphs, they have two major limitations. First, they capture only pairwise pixel relationships, making it difficult to represent and aggregate higher order, multipixel correlations; second, their computational cost is substantial, rendering them unsuitable for direct application to large-scale remote sensing data. In contrast,

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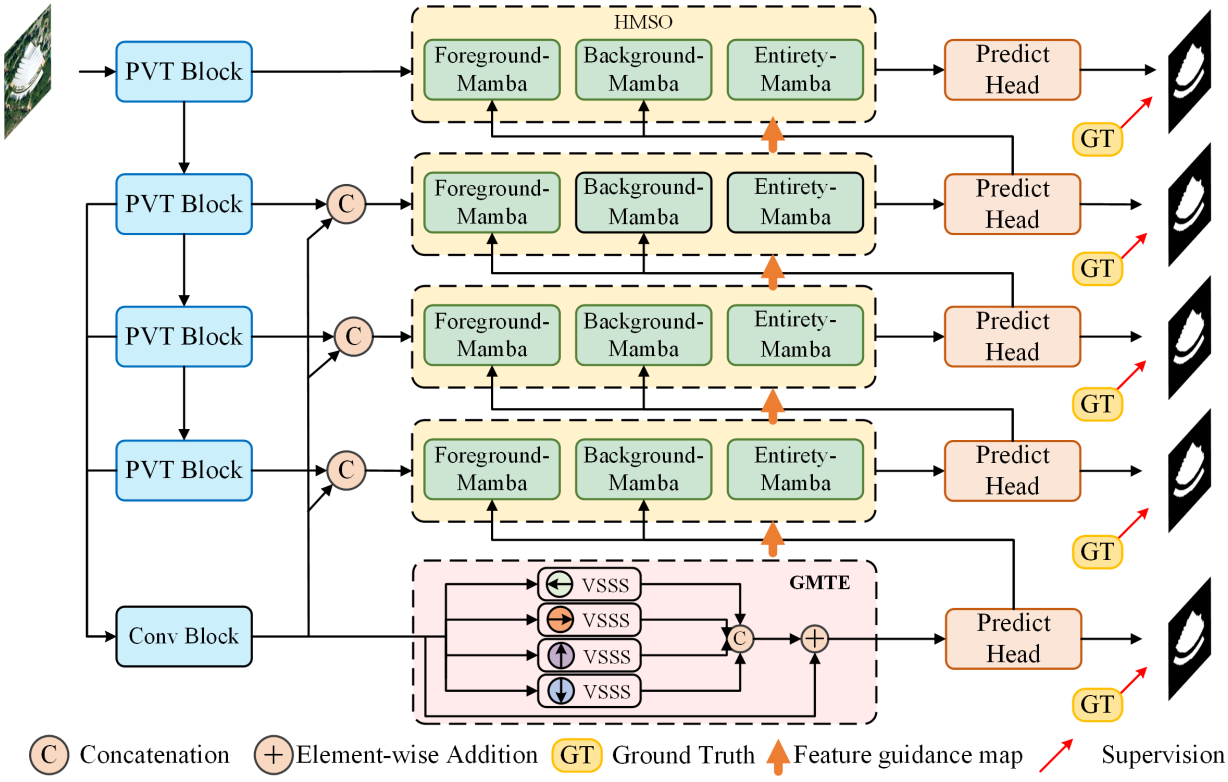


Fig. 2. Structure of THMNet.

conventional CNNs, while computationally efficient, are constrained by limited local receptive fields, which hinders their ability to effectively capture object shape-topology and global semantic relationships between foreground and background. To address these challenges, we propose a GMTE module, which aims to improve global modeling capability while maintaining computational efficiency, thereby optimizing object shape-topological features. Specifically, GMTE employs four independent Mamba structures, each scanning feature maps along a different direction. During this process, Mamba integrates accumulated contextual information to comprehensively model complex many-to-many semantic relationships. Given the small target boundary sizes and significant background interference in remote sensing images, directional modeling aggregates local pixel information along multiple orientations, reinforcing boundary continuity through accumulated context and precisely characterizing object shape topology, thus improving the salient object segmentation accuracy. Moreover, Mamba achieves long-range dependency modeling with linear computational complexity, enabling efficient capture of global topological information. In this manner, GMTE can comprehensively model the semantic relationships between foreground and background while further enhancing object shape topology, ultimately improving overall detection performance.

As illustrated in Fig. 3, the core design of the GMTE module lies in balancing computational efficiency with topological modeling capability. It effectively integrates contextual information across multiple scales, capturing global structural patterns and cross-scale correlations of the input feature maps, thereby providing more discriminative shape priors for subsequent segmentation. To achieve this, GMTE employs group-based Mamba modules to capture the global contextual relationships present in the image. Specifically, the GMTE module receives a deep feature map as input and divides it into four subgroups along the channel dimension, with each subgroup containing one-fourth of the original feature map's channels. Each group of feature maps is then processed through an independent visual single selective scanning block (VSSS Block), so as to extract rich spatial and directional information. To better model the spatial dependencies and global topological relationships of the image, each VSSS Block computes using one of four distinct scanning directions: left-to-right, right-to-left, bottom-to-top, and top-to-bottom. Through these multidirectional scanning approaches, the VSSS Block can analyze the input feature map from various perspectives, fully capturing the global spatial dependencies and contextual information within the image. In Fig. 3, the scanning paths for each direction are detailed in the direction, where each pixel block is processed independently in all four directions, progressively establishing a complete global relationship. The advantage of this grouping strategy lies in its significant reduction of computational complexity, enabling a lightweight design while comprehensively covering the global information across the spatial dimensions of the image. This enhances the model's ability to accurately distinguish between the foreground and background, particularly in complex environments. Through the modeling of multidirectional spatial

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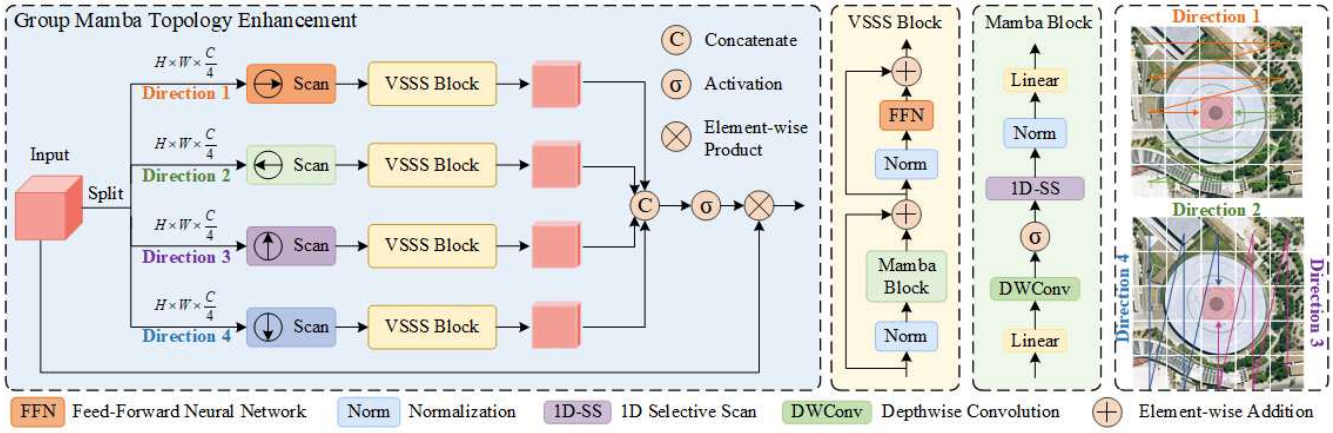


Fig. 3. Structure of GMTE Module.

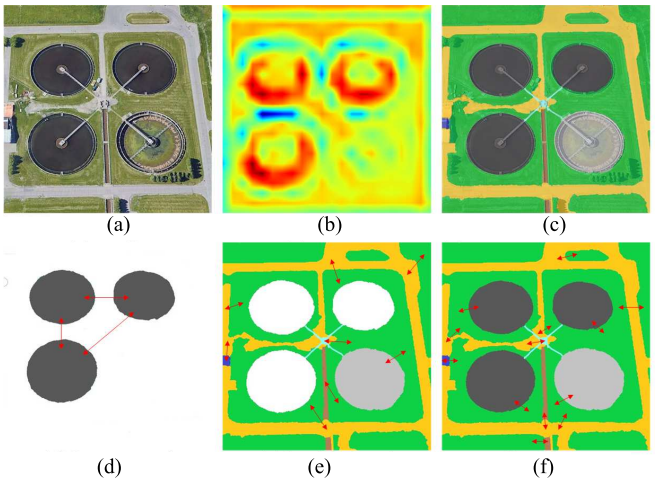


Fig. 4. (a) Image. (b) Feature maps output by the encoder. (c) Visualization of the prior information after object shapes are refined by the GMTE module. (d)–(f) Visualizations of object information in the foreground, background, and overall scene processed by HMSO, respectively, with red arrows being spatial topological connections between objects.

dependencies, the GMTE module excels in capturing long-range dependencies, thereby further improving the accuracy and efficiency of SOD in remote sensing images.

Here, $G = 4$ is the number of groups for four directions: left-to-right, right-to-left, bottom-to-top, and top-to-bottom. The GMTE module is designed to operate with four directional groups. The input feature map X_{inout} is divided into four sequences, each denoted by x_{LR}, x_{RL}, x_{BT}, x_{TB}, with uniform dimensions $(H \times W \times C/4)$. Assuming that the directional parameters fed into the VSSS Block for these four groups are represented as θ_{LR}, θ_{RL}, θ_{BT}, and θ_{TB}; then, the overall relationship for the grouped Mamba can be expressed as follows:

$$\begin{cases} X_{\text{LR}}, X_{\text{RL}}, X_{\text{BT}}, X_{\text{TB}} = \text{Split}\left(X_{\text{input}}\right) \\ \text{GroupMamba}\left(X_{\text{input}}\right) = \text{concat}\left(\text{VSSS}\left(X_{\text{LR}}, \theta_{\text{LR}}\right)\right. \\ \text{VSSS}\left(X_{\text{RL}}, \theta_{\text{RL}}\right) \\ \text{VSSS}\left(X_{\text{BT}}, \theta_{\text{BT}}\right) \\ \left.\text{VSSS}\left(X_{\text{TB}}, \theta_{\text{TB}}\right)\right) \\ Y = \left(\text{act}\left(\text{GroupMamba}\left(X_{\text{input}}\right)\right) * X_{\text{input}}\right) \end{cases} \tag{1}$$

where split() is the splitting operation, concat() is the concatenation along the channel dimension, act() is the activation function, * indicates the dot multiplication, and Y represents the final output. The VSSS Block receives feature maps and directions and performs feature enhancement. Specifically, upon receiving the grouped feature map, the VSSS Block initiates feature enhancement by first normalizing the feature map. This normalization step ensures consistency in feature scales for subsequent extraction processes. The normalized feature map is then processed through the Mamba module, which executes the selective scanning feature extraction operation. This selective scanning process focuses on significant information within the feature map by employing various scanning patterns, thereby suppressing irrelevant areas and enhancing the ability to capture global context. Following the selective scanning feature extraction, the output features are connected to the input feature map through a residual connection. This connection preserves the flow of information and helps avoid feature degradation. Next, the feature map undergoes normalization once more to ensure a consistent distribution of features. The normalized feature map is subsequently passed through a feedforward neural network (FFN) to further extract and reinforce local features. The outputs from the FFN are again connected to the prior feature map through a residual connection, ultimately yielding a rich, multilayered feature representation. The processing flow can be expressed as follows:

$$\begin{cases} X_1 = \text{Mamba}\left(\text{Norm}\left(X\right)\right) + X \\ Y = \text{FFN}\left(\text{Norm}\left(X_1\right)\right) + X_1 \end{cases} \tag{2}$$

where X is the input, Y is the output, Mamba() is the Mamba Block, Norm() is the normalization operation, and FFN() is the feedforward neural network. Through multidirectional selective scanning, the VSSS Block captures spatial dependencies from various directions while employing grouped processing to reduce computational cost. This module introduces selective enhancement in feature extraction, allowing it to filter out key information and suppress redundancy. In addition, residual connections are utilized to retain the original features, ensuring smooth information flow. The hierarchical feature learning approach of this module is particularly well-suited for

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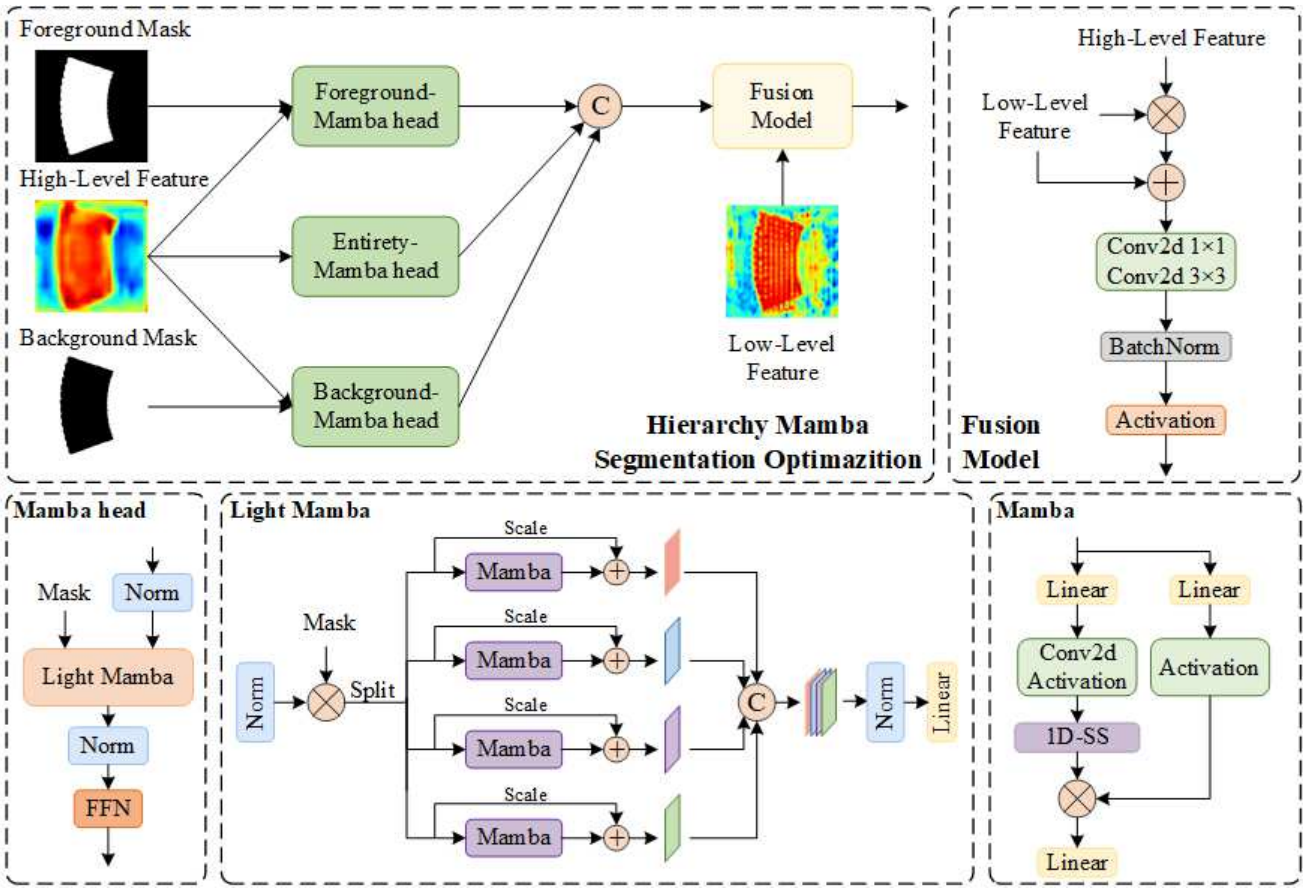


Fig. 5. Structure of the HMSO module.

high-accuracy SOD tasks in complex backgrounds, achieving efficient computational performance while maintaining robust global information modeling capabilities. In summary, GMTE effectively reinforces the global semantic and directional structural information of feature maps through multidirectional modeling, thereby providing high-quality, topologically constrained features for subsequent SOD.

C. Hierarchy Mamba Segmentation Optimization

SOD in remote sensing imagery often poses challenges when it comes to separating foreground and background in complex scenes. In high-resolution images, the background information can be cluttered, and the features of the foreground and background often exhibit significant similarities, making accurate differentiation difficult. In addition, the irrelevant information in complex backgrounds can interfere with the detection of salient objects, adversely affecting the overall detection accuracy. Many existing methods, when addressing the differentiation between foreground and background, tend to rely on highly coupled modules that indiscriminately process features from both foreground and background. They lack mechanisms specifically designed to enhance either foreground or background features. This approach fails to fully leverage the decoupled multibranch structure to effectively emphasize the contrast between foreground and background, and it is also limited in capturing the spatial topological relationships among objects as well as between foreground and background targets. Consequently, it often yields suboptimal segmentation results in complex backgrounds. To address this, we propose the HMSO module, which centers on three decoupled Mamba heads. Using high-level feature maps and mask maps as prior information, these heads separately refine and enhance foreground, background, and overall features, modeling the spatial topological relationships among targets and enabling hierarchical feature optimization and deep coordination. As illustrated in Fig. 4, the GMTE module receives multiscale features from the encoder and integrates them at the global level to achieve fine-grained shape-topology modeling.Its primary objective is to capture internal connectivity and boundary continuity of targets, thereby producing high-quality shape priors. Guided by these priors, the HMSO module further focuses on the spatial topological relationships among objects. Specifically, HMSO partitions features into foreground, background, and global levels, modeling spatial relations within each region using decoupled Mamba structures. The foreground branch emphasizes the topological relationships within foreground regions to reduce omission of salient targets; the background branch reinforces topological features in background regions to suppress interference, and the global branch characterizes complex topological interactions between foreground and background from an overall perspective. Through this hierarchical spatial-topology optimization, HMSO leverages

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GMTE-provided shape priors to achieve multilevel enhancement from foreground and background to the overall scene, yielding more robust and precise segmentation results in complex remote sensing scenarios.

The HMSO module uses high-level feature maps and mask maps as guiding priors to fine-tune the features at each spatial location. Specifically, the multiscale feature maps output by the encoder are first uniformly transformed into 128-D features through a 1×1 convolutional module. Within the decoder, a progressive strategy is employed to refine features layer-by-layer, starting from the top of the encoder. The HMSO conducts optimization at each feature level and generates masks through detection heads. These masks serve a dual purpose: they act as guiding priors for subsequent layers and apply side supervision to the mask map at each feature level to enhance the guidance accuracy. This approach enables the model to more effectively process and distinguish salient objects from the background.

As illustrated in Fig. 5, when the HMSO module receives high-level feature maps and mask maps as input, it initially employs three Mamba heads to focus on enhancing the topology of the foreground, background, and overall regions separately. The outputs of these three modules are then concatenated along the channel dimension and passed through a fusion module for collaborative feature learning. This process captures the differences and correlations between the foreground, background, and overall features, thereby strengthening the comprehensive understanding of salient objects. When the high-level feature map is denoted as X_{high}, the low-level feature map as X_{low}, and the mask map as X_{Mask}, the processing of HMSO can be expressed as follows:

$$\begin{aligned} X_F &= \text{FgHead}\left(X_{\text{high}}, X_{\text{Mask}}\right) \\ X_B &= \text{BgHead}\left(X_{\text{high}}, 1 - X_{\text{Mask}}\right) \\ X_E &= \text{EtHead}\left(X_{\text{high}}\right) \\ Y &= \text{Fusion}\left(X_{\text{low}}, \text{concat}\left(X_F, X_B, X_E\right)\right). \end{aligned} \tag{3}$$

In this process, the feature maps are first normalized and then input into the Light Mamba together with the corresponding mask maps. The Light Mamba significantly reduces the module's dependency on the number of input channels through parallel processing. For example, given a feature map with C channels, Light Mamba splits it equally into four parts along the channel dimension, each with a number of channels of $C/4$, denoted by $X_1^{C/4}$, $X_2^{C/4}$, $X_3^{C/4}$, and $X_4^{C/4}$. Each of these segmented features is then processed by the corresponding submodules of the Light Mamba block, which enhances the modeling of long-range spatial dependencies. The outputs from these modules are combined with the input feature maps using residual connections, and a residual coefficient is applied to optimize feature transfer. Finally, the processed feature maps are reconcatenated along the channel dimension, followed by normalization and a linear layer to generate the final output, further improving the precision and efficiency of feature representation. This operation can be represented mathematically as follows:

$$\begin{cases} X_1^{C/4}, X_2^{C/4}, X_3^{C/4}, X_4^{C/4} = \text{Split}\left(\text{Norm}\left(X_{\text{in}}^{C}\right) * \text{Mask}\right) \\ \text{Mam}X_i^{C/4} = \text{Mamba}\left(X_i^{C/4}\right) + \theta \cdot X_i^{C/4} \quad i = 1, 2, 3, 4 \\ X_{\text{out}} = \text{concat}\left(\text{Mam}X_1^{C/4}\right. \\ \text{Mam}X_2^{C/4} \\ \text{Mam}X_3^{C/4} \\ \left.\text{Mam}X_4^{C/4}\right) \\ Y = \text{Linear}\left(\text{Norm}\left(X_{\text{out}}\right)\right) \end{cases} \tag{4}$$

where X_{in}^{c} is the input, Y is the output, concat() denotes concatenation along the channel dimension, Norm() represents normalization, Linear() refers to the linear layer, and θ is the learnable residual coefficient. Through parallel Mamba processing of the features, the system maintains the number of channels while minimizing the parameter count and preserving high accuracy.

HMSO leverages the refined shape-topology information provided by GMTE to delineate the contours of individual objects. Building on this, it performs decoupled, focused modeling of foreground, background, and overall features to further refine the spatial topological relationships among objects. This approach effectively suppresses background interference, enabling the network to more accurately capture the boundaries and morphology of salient targets in complex scenes.

D. Hybrid Loss

During the training process, a mixed loss is defined to facilitate the backpropagation and updating of model parameters. The overall loss consists of a pixel-level loss, BCE, and a map-level loss, IoU, which can be formulated as follows:

$$L_{\text{total}} = \sum_{i=1}^{4} \left(\lambda_1 L_{\text{BCE}}\left(P_i, \text{GT}\right) + \lambda_2 L_{\text{IoU}}\left(P_i, \text{GT}\right)\right) \tag{5}$$

where GT is the true saliency label graph and λ_1 and λ_2 are the weight coefficients for the BCE and IoU losses, with default values set to 1 to control their relative contributions.

IV. Experiments

A. Datasets

To evaluate the performance of THMNet, two publicly available datasets were used for SOD in remote sensing images: ORSSD and EORSSD. Each dataset comprises original images and their corresponding pixel-level labels. Notably, to enhance the training efficacy of the model, an eightfold data augmentation was applied to the training samples. Below is a detailed description of the three datasets.

1) ORSSD [17]: This dataset, first released by Li et al., is a publicly available collection for SOD in remote sensing images, containing a total of 800 images. Out of these, 600 images were designated for training, while the remaining 200 were for testing. The images are primarily sourced from remote satellite and aerial systems via Google Earth, with a small portion obtained from other remote sensing image

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Fig. 6. EORSSD complex-scene testing images. (a) Background exhibits clutter with appearance and texture similar to the foreground. (b) Image features an intricate spatial structure.

databases. The spatial resolution of the images within this dataset is 0.5–2 m, and the resolution ranges from 256 × 256 to 2107 × 1198. ORSSD contains eight categories of salient objects, including airplanes, ships, cars, rivers, ponds, bridges, stadiums, and beaches.

2) EORSSD [18]: As an extension of ORSSD, EORSSD introduces an additional 1200 images, also sourced from Google Earth. It contains a total of 2000 samples, with 1400 allocated for training and 600 reserved for testing. EORSSD presents heightened complexity and challenges compared with ORSSD. First, images containing multiple salient objects account for 36.5% of EORSSD, thereby increasing the difficulty of multiobject detection. Second, EORSSD includes scenes with small objects (where the object occupies less than 10% of the image) in a staggering 84.65% of the cases, further complicating the detection of small targets.

3) EORSSD Complex-Scene: To more comprehensively validate the effectiveness of the model in complex-background scenarios, we constructed a representative subset from the EORSSD testing set. As shown in Fig. 6, some sample images from this subset are presented. Specifically, images were selected based on the following criteria: 1) the background contains a substantial amount of interference information with appearances or textures similar to the foreground [see Fig. 6(a)] and 2) the image contains multiple salient targets or exhibits complex spatial layout relationships [see Fig. 6(b)]. Based on these criteria, a total of 129 images were selected to form the complex-background subset, which was used to further evaluate the robustness and generalization capability of the detector in challenging scenarios.

B. Experimental Details

1) Parameter Settings: This study was conducted using the PyTorch framework on high-performance GPUs (NVIDIA RTX A100) equipped with 80GB of VRAM. During training, PVTv2 was used as the feature extraction network and initialized using pretrained weights. The input image size was set to 352 × 352, and the Adam optimizer was employed. The batch size was set to 32 and the total number of training epochs was set to 60. The initial learning rate was 1×10^{-4} and was adjusted to 1×10^{-5} after the 45th epoch to enhance convergence.

2) Evaluation Metrics: To comprehensively evaluate the model performance, several metrics are employed for quantitative comparisons with different methods. The S-measure [40] assesses the structural similarity between predicted results and ground truth from both the target and background, with the parameter α set to 0.5 to balance the importance of the target and background. The mean absolute error (MAE) is used to measure the absolute error between the predictions and the actual labels, serving as a crucial indicator of prediction accuracy, with lower values being preferable. The E-measure [41] focuses on the differences between global averages and local pixels, evaluating the alignment of overall features and local details in the results. The F-measure [42] evaluates the quality of the saliency maps by balancing precision and recall. Among these metrics, a lower MAE indicates smaller model errors, while higher values for the other indicators reflect better model performance.

C. Comparison With State-of-the-Art Methods

This study conducts both visual and quantitative comparisons of the proposed method against 21 competitive SOD methods. These methods can be categorized into convolution-based methods (including SAMNet [43], HVPNet [44], DAFNet [18], MSCNet [45], MJRBM [8], PAFR [46], CorrNet [47], EMFINet [7], MCCNet [6], ACCoNet [48], AESINet [22], ERPNet [19], ADSTNet [49], SFANet [25], and LiteSalNet [50]), and Transformer-based methods (including VST [51], ICON [52], HFANet [53], TLCKDNet [54], PEFFNet [55], and RAMENet [56]).

1) Visual Comparison: To validate the superiority of our method (denoted by Ours) in visual detection performance, Fig. 7 presents the visual results of Ours compared with existing methods across various challenging scenarios. Specifically, we selected a series of representative complex scenes characterized by detection challenges, including those affected by intricate backgrounds, scenarios where the background topology closely resembles the target, cases with low contrast between the target and background, and instances featuring complex-background structures. The visual performance in these scenarios shows that our method has a significant advantage as it can effectively capture and display essential details within the images.

The detection is affected by the complex background. In scenarios where the detection is affected by intricate backgrounds, similar textures and shapes in the background can adversely impact detection accuracy. As shown in the two examples in Fig. 7, circular and semicircular buildings are surrounded by complex-background textures, making it difficult to distinguish the edges of the targets. Other networks, such as SAMNet and MSCNet, perform poorly under such conditions, struggling to accurately distinguish between foreground and background. For instance, SAMNet produces blurred contours when detecting buildings in the first row, while MSCNet suffers from missed detections under background interference. In contrast, our method effectively captures the long-range topological relationships between the targets and the background, enabling the model to display target contours clearly even amid complex backgrounds. This significantly reduces false positives and missed detections caused by background interference, thus maintaining both the integrity and clarity of the detections.

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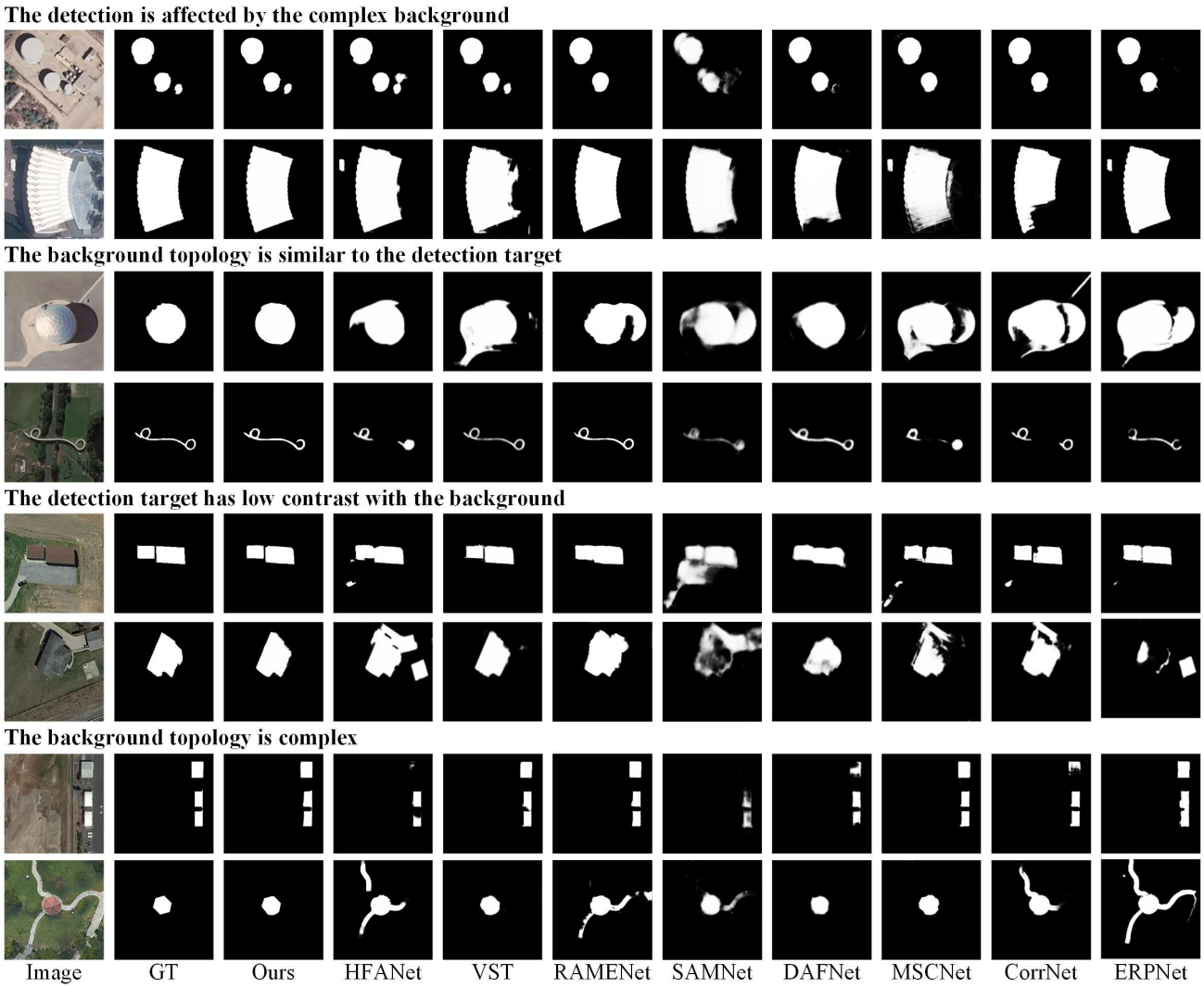


Fig. 7. Visual results of our method and other methods in different scenarios.

The background topology is often similar to that of the detection targets: making separation particularly challenging. As illustrated in Fig. 7, when objects with shapes similar to buildings coexist in the background, other networks such as DAFNet and MSCNet tend to produce blurry segmentation results, where the foreground and background are difficult to distinguish and object boundaries remain unclear. Specifically, DAFNet exhibits obvious cases of missed detection, while MSCNet incorporates background information into the target regions, leading to inaccurate segmentation. In contrast, our method refines and enhances the topology of both targets and background across multiple levels, thereby improving the boundary discrimination and maintaining the integrity of target contours even under similar background interference. This results in more accurate and clearer segmentation outcomes.

The detection target has low contrast with the background: in such scenarios, the minimal difference in color or brightness between the target and background often leads to boundary confusion and a substantial drop in segmentation accuracy. As shown in the low-contrast examples in Fig. 7, other networks such as RAMENet and ERPNet frequently produce segmentation results with blurred contours or missed detections, making it difficult to clearly separate the target from the background. Specifically, RAMENet struggles to capture complete object contours under low contrast, resulting in outputs with obvious boundary blurring, while ERPNet tends to overlook small details in low-contrast regions, leading to partial target omission. In contrast, our method successfully captures more subtle details and distinguishes targets from background, thereby achieving clear and accurate segmentation even under low-contrast conditions. This advantage stems from the ability of our modules to effectively fuse local detail with global structural information, ensuring that the overall object morphology is preserved while avoiding the blurring and missed detections typically caused by insufficient contrast.

The background topology is complex: In such scenarios, our method exhibits superior target highlighting capabilities, accurately segmenting targets despite intricate background interference. Such scenes typically contain considerable background details and complex structures that can affect the

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TABLE I
QUANTITATIVE COMPARISONS WITH STATE-OF-THE-ART SOD METHODS ON EORSSD DATASETS

Methods	EORSSD							
	S_a	MAE	E_ξ^{adp}	E_ξ^{mean}	E_ξ^{max}	F_β^{adp}	F_β^{mean}	F_β^{max}
CNN-based SOD methods								
SAMNet	0.8622	0.0132	0.8284	0.8700	0.9421	0.6114	0.7214	0.7813
HVPNet	0.8734	0.0110	0.8270	0.8721	0.9482	0.6202	0.7377	0.8036
DAFNet	0.9166	0.0060	0.8443	0.9290	0.9859	0.6423	0.7842	0.8612
MSCNet	0.9071	0.0090	0.9329	0.9551	0.9689	0.7553	0.8151	0.8539
MJRBM	0.9197	0.0099	0.8897	0.9350	0.9646	0.7066	0.8239	0.8656
PAFR	0.8927	0.0119	0.8959	0.9210	0.9490	0.7123	0.7961	0.8260
CorrNet	0.9289	0.0083	0.9593	0.9646	0.9696	0.8311	0.8620	0.8778
EMFINet	0.9319	0.0075	0.9500	0.9598	0.9712	0.8036	0.8505	0.8742
MCCNet	0.9327	0.0066	0.9538	0.9685	0.9755	0.8137	0.8604	0.8904
ACCoNet	0.9290	0.0074	0.9450	0.9653	0.9727	0.7969	0.8552	0.8837
AESINet	0.9358	0.0079	0.9462	0.9636	0.9751	0.7923	0.8524	0.8838
ERPNet	0.9210	0.0089	0.9228	0.9401	0.9603	0.7554	0.8304	0.8632
ADSTNet	0.9311	0.0065	0.9681	0.9709	0.9769	0.8532	0.8716	0.8804
SFANet	0.9349	0.0058	0.9669	0.9726	0.9769	0.8492	0.8680	0.8833
LiteSalNet	0.9273	0.0063	0.9729	0.9634	0.9743	0.8796	0.8809	0.8877
Transformer-based SOD methods								
VST	0.9208	0.0067	0.8941	0.9442	0.9743	0.7089	0.8263	0.8716
ICON	0.9185	0.0073	0.9497	0.9619	0.9687	0.8065	0.8371	0.8622
HFANet	0.9380	0.0070	0.9644	0.9679	0.9740	0.8365	0.8681	0.8876
TLCKDNet	0.9350	0.0056	0.9514	0.9661	0.9788	0.7969	0.8535	0.8843
PEFFNet	0.9387	0.0066	0.9664	0.9746	0.9827	0.8384	0.8707	0.8942
RAMENet	0.9419	0.0050	0.9735	0.9778	0.9822	0.8678	0.8840	0.8971
Ours	0.9431	0.0054	0.9722	0.9771	0.9826	0.8647	0.8820	0.8980

The best result is in red, and the second-best is in blue, and the third-best is in green.

network's ability to recognize targets, leading to false positives and misclassifications. Observational results indicate that other networks (such as ERPNet and CorrNet) often misidentify background areas as targets under complex backgrounds or display confusion at target boundaries, failing to achieve the precise separation of the targets. In contrast, our method performs exceptionally well. This demonstrates that even in the presence of complex topological features in the background, our method can effectively distinguish between foregrounds and backgrounds while accurately segmenting salient objects.

2) Quantitative Comparison: According to the quantitative comparison results from Tables I and II on the EORSSD and ORSSD datasets, our method performs exceptionally well across multiple metrics. On the EORSSD dataset, our method ranks within the top three across all eight evaluation metrics. This indicates that the feature enhancement strategy utilizing decoupled heads significantly improves detection effectiveness, enabling the model to more precisely locate salient objects. In addition, the introduction of Mamba demonstrates considerable potential in handling segmentation tasks involving complex backgrounds and salient targets, confirming its effectiveness in SOD. Furthermore, in comparisons on the ORSSD dataset, our method consistently ranks among the top across all eight evaluation metrics, further demonstrating the stability and generalization capability of our model. In contrast, CNN-based methods such as DAFNet and ACCoNet, while performing well on certain metrics, generally struggle to match the overall performance of our model. These methods are prone to interference, particularly in the case of segmentation boundaries from complex backgrounds. Furthermore, Transformer-based methods like VST and ICON, despite having some advantages in modeling long-range dependencies, still fall short in preserving salient details and boundary accuracy. This shortcoming may be ascribed to the Transformer architecture's inadequate focus on details in SOD.

To validate the effectiveness of our method in complex scenarios, we compared its performance with mainstream approaches on the EORSSD complex-scene subset. As shown in Table III, the results demonstrate that our method achieves superior performance in challenging scenes, consistently ranking among the top across all evaluation metrics. The exceptional performance of our model can largely be attributed to the hierarchical topological enhancement capabilities of the Mamba module. In SOD tasks, issues related to complex backgrounds and low-contrast environments often present significant challenges. Other networks often have missed detections and false positives in these scenarios; however, our model effectively addresses these issues through feature

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TABLE II
QUANTITATIVE COMPARISONS WITH STATE-OF-THE-ART SOD METHODS ON ORSSD DATASETS

Methods	ORSSD							
	S_a	MAE	E_ξ^{adp}	E_ξ^{mean}	E_ξ^{max}	F_β^{adp}	F_β^{mean}	F_β^{max}
CNN-based methods								
SAMNet	0.8761	0.0217	0.8656	0.8818	0.9478	0.6843	0.7531	0.8137
HVPNet	0.8610	0.0225	0.8471	0.8717	0.9320	0.6726	0.7396	0.7938
DAFNet	0.9191	0.0113	0.9360	0.9539	0.9771	0.7876	0.8511	0.8928
MSCNet	0.9227	0.0129	0.9584	0.9653	0.9754	0.8350	0.8676	0.8927
MJRBM	0.9204	0.0163	0.9328	0.9415	0.9623	0.8022	0.8566	0.8842
PAFR	0.8938	0.0211	0.9315	0.9268	0.9467	0.8025	0.8275	0.8438
CorrNet	0.938	0.0098	0.9721	0.9746	0.979	0.8875	0.9002	0.9129
EMFINet	0.9432	0.0095	0.9715	0.9726	0.9813	0.8797	0.9000	0.9155
MCCNet	0.9437	0.0087	0.9735	0.9758	0.9800	0.8957	0.9054	0.9155
ACCoNet	0.9437	0.0088	0.9721	0.9754	0.9796	0.8806	0.8971	0.9149
AESINet	0.9460	0.0086	0.9707	0.9747	0.9828	0.8666	0.8986	0.9183
ERPNet	0.9254	0.0135	0.9520	0.8566	0.9710	0.8356	0.8745	0.8974
ADSTNet	0.9379	0.0086	0.9785	0.9740	0.9807	0.8979	0.9042	0.9124
SFANet	0.9453	0.0070	0.9765	0.9789	0.9830	0.8984	0.9063	0.9192
LiteSalNet	0.9372	0.0090	0.9752	0.9700	0.9789	0.9056	0.9061	0.9124
Transformer-based methods								
VST	0.9365	0.0094	0.9466	0.9621	0.9810	0.8262	0.8817	0.9095
ICON	0.9256	0.0116	0.9554	0.9637	0.9704	0.8444	0.8671	0.8939
HFANet	0.9399	0.0092	0.9722	0.9712	0.9770	0.8819	0.8981	0.9112
TLCKDNet	0.9421	0.0082	0.9696	0.9710	0.9794	0.8719	0.8947	0.9114
PEFFNet	0.9372	0.0059	0.9848	0.9857	0.9901	0.9003	0.9141	0.9272
RAMENet	0.9489	0.0070	0.9797	0.9819	0.9862	0.9097	0.9147	0.9229
Ours	0.9503	0.0071	0.9775	0.9798	0.9843	0.9023	0.9124	0.9251

The best result is in red, and the second-best is in blue, and the third-best is in green.

TABLE III
QUANTITATIVE COMPARISONS WITH STATE-OF-THE-ART SOD METHODS ON EORSSD COMPLEX-SCENE DATASETS

Methods	EORSSD complex-scene							
	S_a	MAE	E_ξ^{adp}	E_ξ^{mean}	E_ξ^{max}	F_β^{adp}	F_β^{mean}	F_β^{max}
TLCKDNet	0.9232	0.0063	0.9547	0.9650	0.9770	0.8177	0.8739	0.9058
LiteSalNet	0.9241	0.0066	0.9815	0.9678	0.9805	0.9133	0.9118	0.9189
HFANet	0.9250	0.0095	0.9632	0.9640	0.9702	0.8476	0.8801	0.9047
CorrNet	0.9214	0.0086	0.9609	0.9651	0.9699	0.8439	0.8795	0.8974
ADSTNet	0.9313	0.0066	0.9748	0.9732	0.9805	0.8766	0.8967	0.9063
AESINet	0.9337	0.0070	0.9440	0.9617	0.9739	0.8103	0.8697	0.9035
RAMENet	0.9324	0.0074	0.9699	0.9706	0.9745	0.8788	0.8979	0.9142
ours	0.9400	0.0055	0.9811	0.9803	0.9852	0.8934	0.9090	0.9266

The best result is in red, and the second-best is in blue.

enhancement and a layer-by-layer optimization strategy. This allows for more accurate separation of targets from the background, enhancing segmentation precision. It is therefore concluded that our decoupled feature enhancement approach, combined with the Mamba module, holds considerable promise in the field of SOD, providing robust technical support for target segmentation in complex environments.

To further validate the superiority of our method, Fig. 8 presents the PR curves and F-measure curves for the ORSSD and EORSSD datasets. The PR curves visually illustrate the relationship between precision and recall; the closer the curve approaches the (1, 1) coordinate, the better the method's performance. From Fig. 8(a) and (c), it is evident that our method's equilibrium point on the PR curve is much closer

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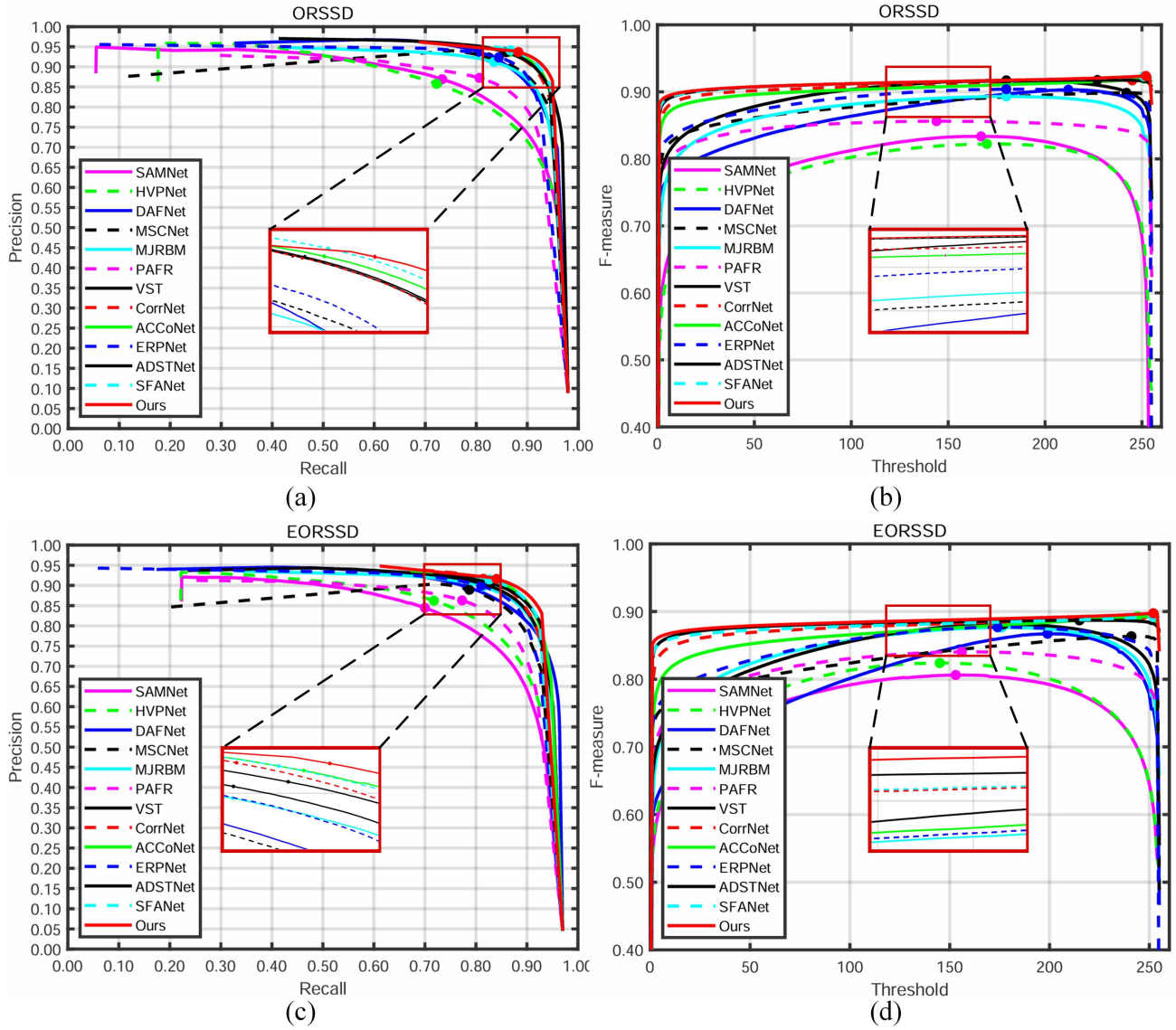


Fig. 8. (a) and (c) PR and (b) and (d) F-measure curves across the two datasets.

to (1, 1) compared with others, demonstrating a clear advantage on both datasets. In addition, in the F-measure curves displayed in Fig. 8(b) and (d), our method consistently outperforms others across various thresholds, showcasing satisfactory performance. These results indicate that our method maintains high detection efficacy across different thresholds, further confirming its stability and superiority.

D. Ablation Studies

1) Contribution Analysis of Independent Modules in THMNet: A series of ablation experiments was conducted on the ORSSD dataset to validate the effectiveness and contributions of various modules within THMNet. In this experiment, four different combinations were designed: 1) a U-Net architecture centered on the PVTv2 encoder as the baseline; 2) Baseline + GMTE; 3) Baseline + HMSO; and 4) THMNet, which integrates all modules (THMNet = Baseline + GMTE + HMSO). The ablation results are presented in Table IV.

As shown in the table, incorporating GMTE into the baseline modeled to increases of 0.51%, 0.84%, 0.83%, and 0.74% across the four metrics on the ORSSD dataset. Furthermore, the addition of HMSO to the baseline resulted in improvements of 0.66%, 0.57%, 0.56%, and 0.41% for the same four metrics. As illustrated in Fig. 9, GMTE enables more precise modeling of object shape features. Compared with the case without GMTE, it produces a clearer delineation of foreground–background boundaries. Furthermore, when modeling the shape-topology information of foreground targets from a global perspective, GMTE simultaneously considers the background topology, thereby providing a solid foundation for the subsequent HMSO module to model spatial topological relationships among different objects. These findings indicate that both the GMTE and HMSO modules significantly enhance the model's performance. In addition, THMNet displayed the best performance across all metrics due to the synergistic effects of the GMTE and HMSO modules. Specifically,

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TABLE IV
CONTRIBUTION ANALYSIS OF EACH MODULE

No.	Method	ORSSD			
		S_α	F_β^{adp}	F_β^{mean}	F_β^{max}
1	Baseline	0.9416	0.8896	0.9015	0.9174
2	Baseline+GMTE	0.9467	0.898	0.9098	0.9248
3	Baseline+HMSO	0.9482	0.8953	0.9071	0.9215
4	THMNet	**0.9503**	**0.9023**	**0.9124**	**0.9251**

The best in each column is in bold.

TABLE V
ABLATION STUDY OF HMSO (ET, FG, AND BG ALL CONTRIBUTE TO THE OVERALL PERFORMANCE)

Settings	HMSO			ORSSD			
	ET	FG	BG	S_α	MAE	F_β^{adp}	F_β^{max}
1				0.9416	0.0096	0.8896	0.9174
2	✓			0.9492	0.0078	0.9008	0.9279
3		✓		0.9449	0.0092	0.8981	0.9218
4			✓	0.949	0.0081	0.8958	0.9221
5		✓	✓	0.9453	0.0086	0.896	0.9214
6	✓		✓	0.9482	0.0085	0.8954	0.92
7	✓	✓		0.9466	0.0087	0.9017	0.9237
8	✓	✓	✓	**0.9503**	**0.0071**	**0.9023**	**0.9251**

The best in each column is in bold.

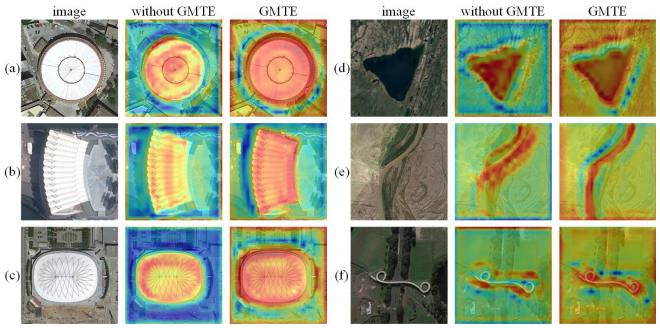


Fig. 9. Visual comparison of feature maps with GMTE ablation. w/o stands for without. (a)–(d) Demonstrate a complex geometric background, while (e) and (f) show elongated salient objects.

THMNet reached 95.03%, 90.23%, 91.24%, and 92.51% in terms of the metrics S_α, F_β^{adp}, F_β^{mean}, and F_β^{max}, respectively.

2) Ablation Study of the Hierarchical Mamba Head: In the following experiments, the contributions of various components within the HMSO module to model performance were analyzed, so as to gain a deeper understanding of how they collectively enhance effectiveness. Table V presents the detailed results of the ablation study, where ET, FG, and BG denote the enhancement mechanisms utilizing "entirety," "foreground," and "background," respectively, during the decoding phase. Setting 1 corresponds to the baseline, which is a simple encoder–decoder structure that aligns with the baseline described in Section IV-D, i.e., the original model without the HMSO module. Experimental results indicate that employing any single enhancement mechanism (ET, FG, or BG) can significantly improve model performance. When using only the ET head, the model achieves strong performance through global context modeling; using only the FG head helps to better capture foreground targets and reduces omission of salient regions; and using only the BG head primarily enhances background understanding and suppresses interference. However, none of these individual branches can match the performance of the complete HMSO. The optimal performance is achieved only when ET, FG, and BG are combined, demonstrating their complementary roles in information modeling: ET provides global structural constraints, FG reinforces foreground capture, and BG improves robustness to complex backgrounds. This set of experiments illustrates that the Mamba segmentation optimization mechanism in the HMSO module effectively refines the targets and backgrounds while integrating global contextual information, thereby significantly elevating model performance. Particularly in the task of SOD, the segmentation of foreground and background helps the model to accurately locate and recognize targets within complex environments. The collaboration among the entirety, foreground, and background enables the HMSO module to optimize feature representation from multiple angles, providing optimal support for salient object segmentation.

V. DISCUSSION

We conducted a comparative analysis of model complexity across multiple networks, including conventional SOD models and other Mamba-based models designed for various vision tasks, using parameters, FLOPs, and FPS as evaluation metrics. Since FPS is not directly comparable across different

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TABLE VI
COMPARISON OF THE MODEL COMPLEXITY

Methods	Params (M)	FLOPs (G)	Speed (fps)
CorrNet	4.086	21.379	61.38
EMFINet	95.086	176	28.03
MCCNet	67.652	114	38.51
ACCoNet	127	50.422	52.54
ERPNet	77.195	171	27.88
GeleNet	25.453	6.43	90.27
RSM-CD [35]	27.9	50.2	-
VMamba [31]	89	15.4	-
Mamba YOLO [37]	21.8	49.7	-
Ours	33.717	13.019	74.25

vision tasks, we report FPS only for SOD models. The results are summarized in Table VI.

As shown in Table VI, our method consistently maintains relatively low parameter counts and computational costs, while also demonstrating advantages in FPS within the SOD task. This indicates that our model achieves superior detection accuracy without introducing excessive redundant computations. Moreover, in comparison with other vision-oriented Mamba models, our method also exhibits lower FLOPs, highlighting its computational efficiency. By leveraging a selective state space and a hardware-aware parallel scanning algorithm, Mamba lowers the core computational complexity of sequence modeling from quadratic to linear. Consequently, the proposed model enables efficient inference and reduces computational burden in practical applications, showcasing both effectiveness and scalability in complex scenarios. These results underscore the strong balance achieved between performance and efficiency, making our approach well-suited even for resource-constrained environments.

Although our method performs favorably among mainstream approaches in overall evaluation metrics, we observe that in certain specific scenarios, such as small-object detection or local detail recovery, some CNN- or Transformer-based methods may still demonstrate competitive performance. For example, in scenes with densely distributed small objects, certain CNN approaches may offer advantages in stability. Conversely, when foreground–background boundaries are extremely blurred, GMTE's ability to capture object shape-topology information may be limited, leading to partial confusion between foreground and background branches. In such ambiguous regions, some Transformer models, leveraging their strong global attention mechanisms, may better capture subtle contrast cues. In future work, we plan to further enhance the model's robustness in small-object and blurred-boundary scenarios, for instance, by incorporating additional topology modeling or uncertainty-guided mechanisms to improve the representation of fine-grained structures in complex scenes.

VI. CONCLUSION

In complex scenes such as remote sensing images, there are often significant topological relationships and irregular spatial structures between the foreground and background. Traditional network architectures struggle to accurately segment salient objects that may be camouflaged or obscured by complex backgrounds. Therefore, this study proposes an SOD network based on topology-aware hierarchical mamba, specifically designed to tackle the difficulties of accurately segmenting salient objects in complex backgrounds. THMNet integrates the GMTE and HMSO modules, utilizing global and hierarchical topology enhancements to refine the modeling of foreground and background features, thereby improving the segmentation accuracy. First, the GMTE module employs grouped Mamba-based topological enhancement to perform global directional modeling on high-level semantic features, with a particular focus on improving the representation of object shape topology, including boundary integrity and internal connectivity. The parallel processing of multiple topological channels enables the model to establish a global understanding of the overall image structure at an early stage of feature extraction, thus laying a solid foundation for subsequent fine-grained segmentation. Guided by the shape priors provided by GMTE, the HMSO module then performs hierarchical focused optimization on foreground, background, and holistic features. This process models the spatial semantic topological relationships among different objects and achieves deep collaboration between foreground and background through multiscale feature feedback, thereby enhancing boundary precision and suppressing background interference. Through the collaborative effect of GMTE and HMSO, THMNet not only optimizes the shape topology of individual objects but also strengthens the spatial semantic topological structures among objects. Overall, this method significantly improves segmentation accuracy and robustness in SOD while enhancing the model's ability to discriminate between foreground and background, demonstrating broad application potential.

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